Question	Answer	Marks
1	**Defining the problem**	
	V is the independent variable and *z* is the dependent variable **or** vary *V* and measure *z*	**1**
	keep *h* constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • pulley supported by stand • stand placed on surface/bench/floor • minimum of two labels from stand, beaker, oil, surface/bench/floor, pulley, string	**1**
	use (metre) rule to measure *h* **or** (metre) rule correctly positioned with *h* marked on diagram	**1**
	use measuring cylinder to measure *V*	**1**
	timing method to measure time *t* of fall of beaker to determine *z* e.g. use timer/stopwatch **or** use light gate(s) connected to a timer/data logger	**1**

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Question	Answer	Marks
1	**Method of Analysis**	
	plot a graph of $\frac{1}{z^2}$ against $\frac{1}{V}$ or equivalent (e.g. $\frac{1}{V}$ against $\frac{1}{z^2}$) Do not accept logarithms.	**1**
	$b = \frac{1}{2h \times y\text{-intercept}}$ (for $\frac{1}{V}$ against $\frac{1}{z^2}$: $b = \frac{M \times \text{gradient}}{ah}$ **or** $b = -\frac{\text{gradient}}{2h \times y\text{-intercept}}$)	**1**
	$a = \frac{M}{bh \times \text{gradient}}$ **or** $a = \frac{2M \times y\text{-intercept}}{\text{gradient}}$ (for $\frac{1}{V}$ against $\frac{1}{z^2}$: $a = -2M \times y\text{-intercept}$)	**1**

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Question	Answer	Marks
1	**Additional detail including safety considerations**	**6**
	D1 precaution linked to oil spillage, e.g. use of cushion/sand box/tray for falling beaker to land **or** use of bungs/lids on beakers **or** use foam on bench/floor **or** use foam to prevent rising beaker hitting pulley	
	D2 precaution linked to oil contact with skin e.g. use gloves to avoid contact with oil	
	D3 keep M constant	
	D4 use a (top-pan) balance to measure M	
	D5 method to keep h constant e.g. use a fiducial mark to release the beaker from the same position **or** release from the same position on the clamped rule each time	
	D6 equation to determine z for method used, e.g. for timing h, $z = 2h / t$ **or** for one light gate, $z = L / t$ where L is the length of the interrupted beam **or** for two light gates, z = distance between light gates / t Do not accept h / t.	
	D7 additional detail on diagram to measure h, e.g. clamp (metre) rule with stand on surface **or** use of set squares positioned on the surface to side of rule **or** spirit level positioned to side of rule	
	D8 use large value of h to increase time of fall of beaker	
	D9 repeat measurements of z for the same V and average z	
	D10 relationship valid if a straight line is produced (passing through $\left(\frac{1}{2bh}\right)$) Do not accept line passing through the origin.	

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